Use PC-PATR with FLEx User Documentation

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1 Introduction

Use PC-PATR with FLEx is a tool that works as a utility in *FieldWorks Language Explorer* (aka *FLEx*). *Use PC-PATR with FLEx* can use a PC-PATR grammar to disambiguate a text or a portion of a text that has been analyzed in a *FLEx* project. You tell *Use PC-PATR with FLEx* the PC-PATR grammar file to use. Then you can choose a text or a portion of that text and ask *Use PC-PATR with FLEx* to try and disambiguate it.

Use PC-PATR with FLEx works with version 9.1.14 Beta or higher of *FLEx* and is only available on 64-bit Windows computers.

While you can use any PC-PATR grammar file, *Use PC-PATR with FLEx* expects that you will have used the *PAWS* program to create the PC-PATR grammar file. See https://software.sil.org/paws/ for more on *PAWS*.

1.1 Invoking *Use PC-PATR with FLEx* from within *FLEx*

While running *FLEx*, use Tools menu item / Utilities.... Find the "Use PC-PATR with FLEx" item, check it, and then click on the "Run Checked Utilities Now" button.

1.2 Initial invocation

The first time you invoke *Use PC-PATR with FLEx* on a *FLEx* database, it will automatically add to your *FLEx* database the following:

1. a custom field to each sense (called "PCPATR") and
2. a custom list (called "PCPATR Feature Descriptors") containing the template names that the PC-PATR grammar generated by the *PAWS* program expects.

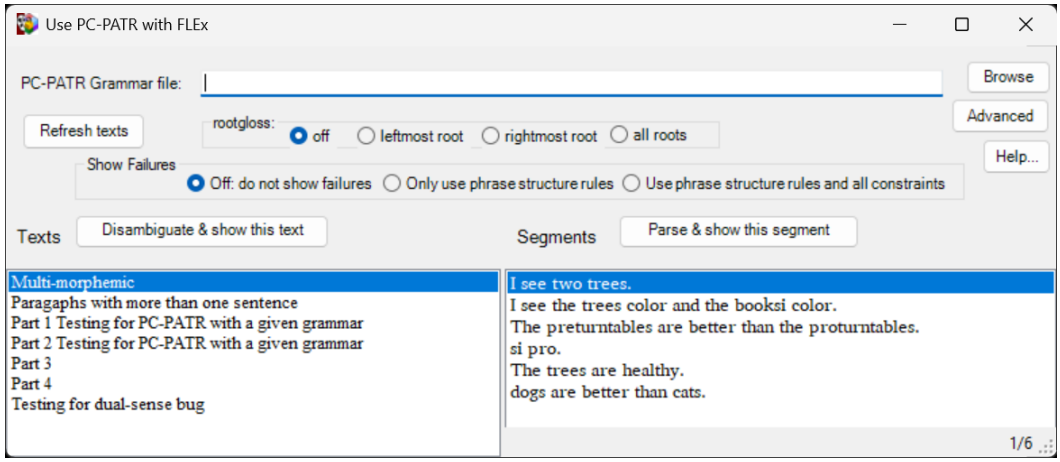
The names shown above are always in the English analysis writing system and English is the only writing system containing these names.

You can find the custom list by clicking on the "Lists" button in *FLEx*.

1.3 Appearance

Use PC-PATR with FLEx looks like what is shown in (1).

(1)



The texts in the *FLEx* database are shown in the left pane and the segments of the first text are shown in the right pane.

There are buttons you can click. Each is discussed in section 2 below.

2 Buttons

You control *Use PC-PATR with FLEx* by using the various buttons. This section briefly describes them. See sections 3–4 for more on what happens for the segment and text buttons, respectively.

2.1 PC-PATR grammar Browse button

To choose which PC-PATR grammar file to use, click on the Browse button. By convention, PC-PATR grammar files have an extension of ".grm" so this is what the file browser uses. **IMPORTANT**: the path to your grammar file (including the grammar file name itself) cannot have any spaces in it. If it does, then *PC-PATR* will not find the file. See below in section 7 for some suggestions on where you can avoid using spaces in the file path.

2.2 Rootgloss choices

PC-PATR allows you to have "rootglosses" in rules. This section allows you to set one of four options with respect to how rootglosses are handled. See section 3.2.14.22 "set rootgloss" in the PC-PATR Reference Manual.[1] The four options are:

1. off: rootglosses are not used; this is the default;
2. leftmost root: the first / leftmost root is used;
3. rightmost root: the last / rightmost root is used;
4. all: every root is used.

2.3 Show Failures choices

Trying to figure out why a given segment fails to parse or why it has extra parses can be facilitated by using the second and third options under "ShowFailures."

1. "Off: do not show failures" is the default. This is for normal, production work.

[1] Use the Help menu item / PC-PATR Reference Manual to see this document.

2. "Only use phrase structure rules" is for when a segment does not parse and you want to see if the phrase structure rules alone produce a parse for the segment. That is, no unification constraints will be applied.[2] Note if that any *FLEx* category is not being correctly mapped to the category used in the PC-PATR grammar, then nothing will show with this option. Make sure every *FLEx* category is correctly mapped to the category used in the PC-PATR grammar file. You can use a "Let" statement to do this.
3. "Use phrase structure rules and all constraints" is to tell *PC-PATR* to use its "show failures" mode. It will report everything it tried while attempting to parse the current segment. Note that this can take a long time so you should only use this when parsing portions of a segment or trying to figure out why a given segment is not parsing at all. Also be aware that if any *FLEx* category is not being correctly mapped to the category used in the PC-PATR grammar, then most likely the output will be a start catgory node with all other categories/words under it in a row. Make sure every *FLEx* category is correctly mapped to the category used in the PC-PATR grammar file. You can use a "Let" statement to do this.

IMPORTANT: when using either the second or third options, you should minimize the number of rules the PC-PATR grammar file uses. See appendix B for a tool that makes this relatively easy to do. If you do not limit the number of rules, *PC-PATR* may well run for a very, very long time. This is why these two options flash red: it is a reminder to you to be careful when using these options.

If by accident you find that *PC-PATR* is running a long time and want to stop it, you can do the following:

1. Start Windows' Task Manager application. One way to do this is to press both the **Ctrl** and **Shift** keys down at the same time, and, while holding them down, press the **Esc** key.
2. Click on the top left icon to show "Processes."
3. At the top, there is a text box showing "Type a name, publisher, or PID to search." Click in it and type "pcpatr64.exe" and it should show "pcpatr64,exe" under "Background processes."
4. Click on "pcpatr64.exe" and then on the "End task" button.
5. The *PcPatr Browser* will then appear but with no sentences or parses showing. Close it and it will take you back to the *Use PC-PATR with FLEx* dialog.

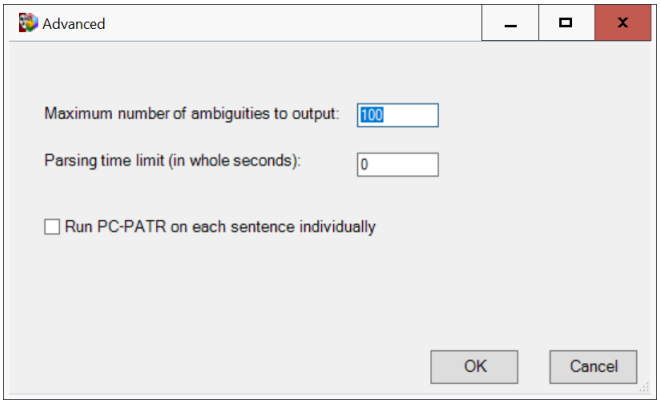
Use PC-PATR with FLEx will remember the settings for each of these three options when you close *Use PC-PATR with FLEx*.

[2]The PC-PATR documentation has this to say about this option: "set unification off turns off feature unification in the grammar. Only the context-free phrase structure rules are used to guide the parse; the feature contraints are ignored. This can be dangerous, as it is easy to introduce infinite cycles in recursive phrase structure rules."

2.4 Advanced

The “Advanced” button is used to changed some advanced options. It brings up a dialog that looks like what is shown in (2).

(2)



There are several options:

1. “Maximum number of ambiguities to output:” is the maximum number of ambiguities that will appear in the *PcPatr Browser*. The default is 100.
2. “Parsing time limit:” is how many seconds to allow for parsing the segments. The default is 0 (which means no time limit).
3. “Run PC-PATR on each sentence individually” is for running each segment individually when disambiguating a text. It is most useful when using a time limit. With this option, you can have *Use PC-PATR with FLEx* parse each segment for a maximum number of seconds given by the parsing time limit. Without this option, the maximum time limit is for the entire text. With this option, the maximum time limit is for each segment.

2.5 Help button

The “Help...” button is used to get this user documentation file, show the “Automatic FLEx to PAWS-generated PC-PATR categories” documentation, or show the PC-PATR Reference Manual.

2.6 Refresh texts button

Whenever you click on the “Refresh texts” button, *Use PC-PATR with FLEx* will reload all of the texts from *FLEx*. This is so if you know that if some texts have been added, deleted or changed since you first started *Use PC-PATR with FLEx*, you can get the most current list of texts.

2.7 Disambiguate button

Above the pane containing the texts is a button labeled “Disambiguate & show this text.” You use this button to try and use the PC-PATR grammar to disambiguate this entire text. See section 4 for more.

2.8 Parse button

Above the pane containing the segments of the selected text is a button labeled "Parse & show this segment." You use this button to try and use the PC-PATR grammar to parse this particular segment. See section 3 for more.

3 Parsing a segment

When you use the "Parse & show this segment" button for a segment or double click on that segment, *Use PC-PATR with FLEx* will pass the parsing information that you entered in *FLEx* to *PC-PATR*, along with the PC-PATR grammar, and process that segment. The result will be shown in a slightly newer version of the *PcPatr Browser* tool (https://software.sil.org/carla/pc-patr-browser/). See appendix A for more information on this tool.

You use this tool to see all the parses the *PC-PATR* program produced for this segment using the PC-PATR grammar file. When there are two or more parses, you can show the correct one and then use Parse / Use This Parse menu item (or click on the ✓ tool bar button) to have this analysis be recorded in the *FLEx* database for this segment. That is, this segment will be disambiguated, using the information from this parse.

If you are using the TreeTran program, you can save the results of the disambiguated parses in a file called Invoker.xml in the %TEMP% directory. You do this by using File / Save for TreeTran to make sure there is a check mark before it. Alternatively, you can click on the item in the tool bar.

When you are done looking at the results, close the *PcPatr Browser* tool and you will return to *Use PC-PATR with FLEx*.

You can also select more than one segment. When you then use the "Parse & show this segment" button, *Use PC-PATR with FLEx* will parse and show all of the selected segments.

4 Disambiguating a text

When you use the "Disambiguate & show this text" button, *Use PC-PATR with FLEx* will pass the parsing information for all segments in the text to *PC-PATR*, along with the PC-PATR grammar, and process all those segments. Since all the segments are processed, this will probably take a while. So please be patient. The result will be shown in the slightly newer version of the *PcPatr Browser* tool. See appendix A for more information on this tool.

Unlike with parsing a segment (see section 3), the result will have as many "sentences" as there are segments in the text.

By default, *Use PC-PATR with FLEx* will disambiguate a segment in the *FLEx* database whenever the result of the parsing uniquely identifies the analysis of each word in the segment. For those cases where there are still ambiguities, you can use the *PcPatr Browser* tool to select a given parse to use. Like with parsing

a segment, you show the correct parse and then use Parse / Use This Parse menu item (or click on the ✓ tool bar button) to have this analysis be recorded in the *FLEx* database for this segment. That is, this segment will be disambiguated, using the information from this parse.

When you are done looking at the results, close the *PcPatr Browser* tool and you will return to *Use PC-PATR with FLEx*.

5 Restarting *Use PC-PATR with FLEx*

Whenever you exit and restart *Use PC-PATR with FLEx*, it will do the following:

1. remember its window size, location, and layout;
2. remember which PC-PATR grammar file you last chose;
3. remember which text in that project you last selected; and
4. remember which segment in that text you last selected.[3]

6 Error messages

In certain situations,*Use PC-PATR with FLEx* will issue an error message.

Table 1 lists the errors *Use PC-PATR with FLEx* reports along with a brief description of what the error might mean.

Error	Meaning
The PC-PATR grammar file had an error in it and failed to load. We will show the error log after you click on OK. Please fix all errors in the grammar file and then try again.	The PC-PATR grammar file being used has at least one error in it so the *PC-PATR* program could not load and use it. The list of errors (and any warning messages, too) will be shown as soon as you click on the OK button.[4] It may show more than one message. Fix the first one and then try again. Often fixing the first error causes the others to go away. Also, the error messages and warnings may be a bit cryptic. See the PC-PATR Reference Manual for (some possible) help on this. (The manual is available via the "Help" button; see section 2.5.)

[3] If you selected more than one segment, *Use PC-PATR with FLEx* will remember the bottom-most one.

[4] Note that in the error message file, the grammar file's location may look odd, but it is the same as what is showing before the “Browse” button. This happens for long file names or folder names. *PC-PATR* is a “DOS” program and uses this cryptic format.

Error	Meaning
The following glosses contain a space. Please fix them and try again.	At least one gloss used in one of the segments has a space in it. This will produce incorrect values in some of the files used. You will continue to get this message until you find and fix the gloss. You should be able to search for the gloss in Lexicon Edit in *FLEx*.
Segment number *number* does not have any words in it. Please remove it or add words and try again.	The indicated segment number does not have any words in it. You need to either remove the segment from the text or add some words to the segment in *FLEx*.
Sorry, but not every sentence has every word parsed. At least the word '*word*' did not parse. Please make sure every word is parsed and try again.	The indicated word does not have a parse. You need to give that word a parse in *FLEx*.
Sorry, but at least the word '*word*' has an analysis with an invalid parse. Please make sure every word has valid parses and try again.	The indicated word does not have a valid parse. You need to fix it in *FLEx*.
The read failed for segment number *number*; *some error message*	Something went wrong while processing the indicated segment number. This only shows when the "Run PC-PATR on each sentence individually" option is set (see section 2.4). It may be that the "Parsing time limit" was met and *Use PC-PATR with FLEx* had some issue trying to read one of its temporary files.

Error	Meaning
The PC-PATR processing failed! Perhaps there are incompatible feature values in one of the forms. We will show the error log after you click on OK. You may also want to try and run the PcPatrFLEx.bat file in the %TEMP% directory. This may or may not show which features are incompatible or some other PC-PATR error or warning message.	When processing the input segment, the PC-PATR grammar file being used has at least one incompatibility in it. The list of errors (and any warning messages, too) will be shown as soon as you click on the OK button. Note that in the error message file, the grammar file's location may look odd, but it is the same as what is showing before the “Browse” button. Also, the error messages and warnings may be a bit cryptic. See the PC-PATR Reference Manual for (some possible) help on this. (The manual is available via the "Help” button; see section 2.5.)

Table 1: *Error messages*

If you get an error message not in the list above, please report it. See section 8.

7 Known problems

The following items are known to be less than desirable with this version of *Use PC-PATR with FLEx*:

1. The path to the grammar file cannot contain any spaces. The *PC-PATR* program will not find the grammar file if there are. One work-around is to put the grammar file in a directory inside a folder at the top level (C:). If your *FLEx* project does not have a space in its name, then you can also try putting the grammar file in “C:\ProgramData\SIL\FieldWorks\Projects*your project name*\SupportingFiles”. The advantage of the latter is that your grammar file can be included in every backup you make of your *FLEx* project and can be included in a Send/Receive operation.
2. *Use PC-PATR with FLEx* assumes that the comment character in the PC-PATR grammar file is a vertical bar |. If you use something else, the parser will fail every segment.
3. When a word in *FLEx* does not have the “Word Cat" field filled out for a word in the interlinear, *Use PC-PATR with FLEx* will try and guess what the category of the word should be. It does this as follows:
   a. If there is only one morpheme, use its category.
   b. If there are two or more morphemes and only one stem/root, find the stem and use its category. (This could be incorrect if one of the affixes is derivational and changes the category.)

   c. If there are two or more stems, use the category of the left-most one. (This could be incorrect if some other stem is the correct category to use or if the stem compounding results in a different category or if one of the affixes is derivational and changes the category.)

   Therefore, there are times you may need to overtly fill out the "Word Cat" field in *FLEx*. Note, however, that this category needs to be the category needed for syntax parsing, not for morphological parsing. For example, if the word is a transitive verb in the syntax, then you need to mark it as a transitive verb and not as a simple verb.
4. When a morpheme in a word analysis has just the form but no sense or grammatical info, then the gloss showing in the output will be the gloss of the first sense of the entry for the morpheme.
5. When a morpheme in a word analysis has the form and grammatical info but no sense overtly indicated and there are two or more senses using the grammatical info in the entry, then the gloss showing in the output will be the gloss of the first sense which uses the grammatical info.
6. The user interface is in English only.

8 Support

If you have any questions with *Use PC-PATR with FLEx* or find bugs in it, please send an email to andy_black@sil.org.

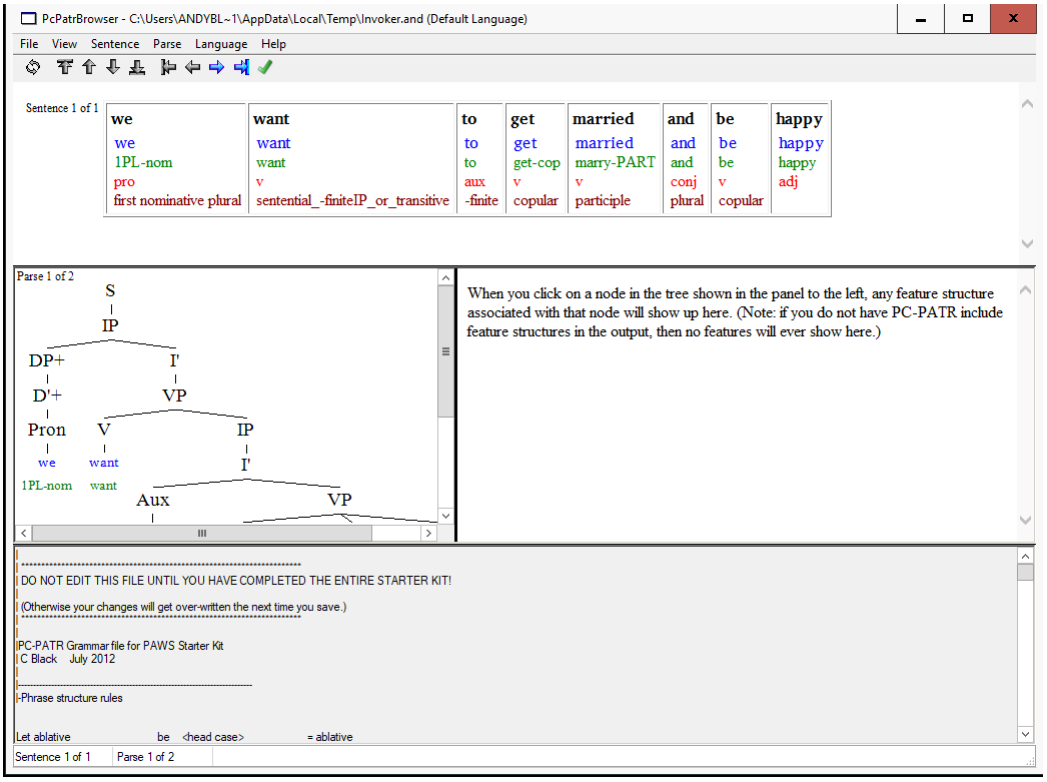
A. The *PcPatr Browser* tool

This appendix briefly covers the various parts of the *PcPatr Browser* tool.

A.1 Overview

Example (3) is a screen shot of what it looks like for one segment using one PC-PATR grammar.

(3)

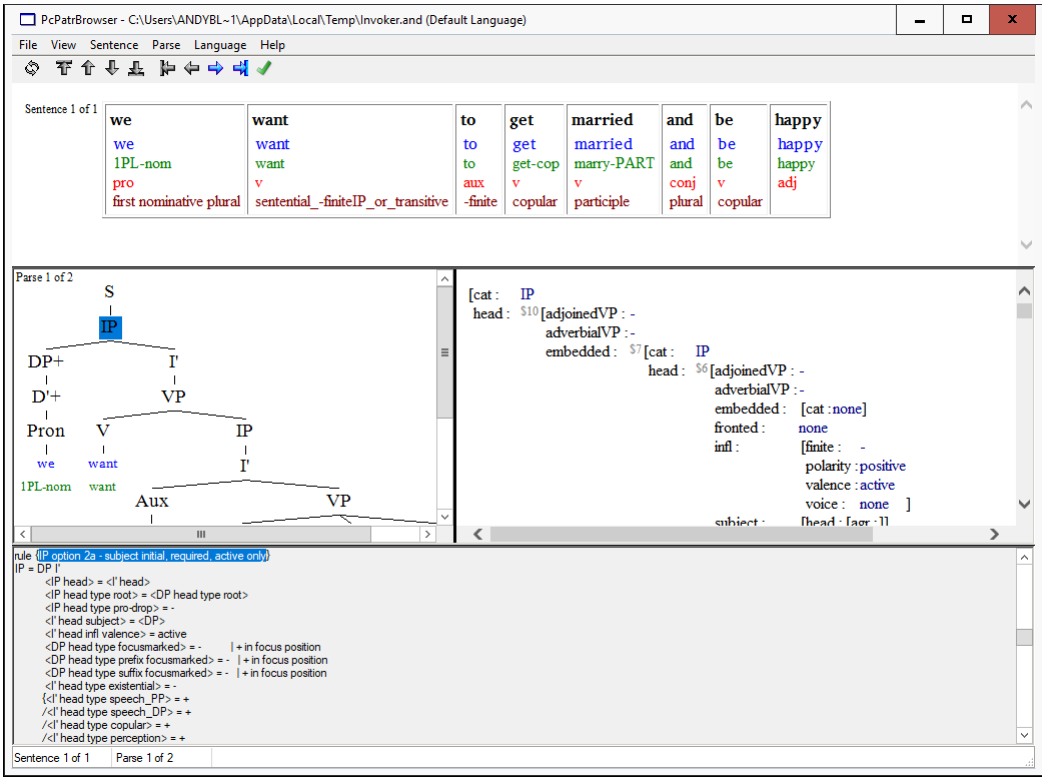


There are four panes:

1. the segment in an interlinear form (at the top);
2. the syntactic tree, if any (left, middle);
3. the feature structure of the selected node in the tree (right, middle); and
4. the PC-PATR grammar file used (bottom)

When a node in the tree is selected, the bottom pane will show the rule in the PC-PATR grammar file that produced that node. In addition, the features for that selected node will appear in the right, middle pane. For example, when clicking on the IP node, it looked like example (4).

(4)



When there is more than one segment, you can use the Sentence menu item options or the first set of arrow buttons on the tool bar to navigate from one sentence to another. See appendix A.3 for keyboard shortcuts. (Please note that with *Use PC-PATR with FLEx*, "sentence" is the same as "segment.")

When there is more than one parse for a given segment, you can use the Parse menu item options or the second set of arrow buttons on the tool bar to navigate from one parse to another. See appendix A.3 for keyboard shortcuts.

When you manually select to use a parse, then the entire tree description will be highlighted in yellow. In addition, the sentence number in the interlinear pane will be highlighted in yellow.

A.2 Right-to-left script

If the vernacular language of your *FLEx* project is a right-to-left script, *Use PC-PATR with FLEx* will try to tell the *PcPatr Browser* this. You may need to use the Language menu item, though, and uncheck and then check the "Use right-to-left orientation in parse tree and interlinear" option.

A.3 Keyboard shortcuts

To make processing texts faster, there are several keyboard shortcuts available. These only work when the "focus" is in the interlinear, tree, or feature panes. When *PcPatr Browser* begins, the default is to have the "focus" on the feature pane. If you have clicked in the *PC-PATR* grammar file pane or if you are using the keyboard

to select a menu item, then these shortcuts will not work. You will need to click in the interlinear, tree, or feature pane first.

The shortcuts are as in example (5):

(5)

Shortcut	Operation
Ctrl + Down Arrow	Next sentence
Ctrl + Up Arrow	Previous sentence
Ctrl + Right Arrow	Next parse
Ctrl + Left Arrow	Previous parse
Ctrl + Shift + Down Arrow	Use this parse & go to next sentence
Ctrl + /[5]	Use this parse
Ctrl + L	Use this parse

B. A good way to edit a PC-PATR grammar file

One challenge with editing a PC-PATR grammar file is that it is a plain text file. If one wants to try and debug a grammar, one has to comment off a large amount of lines. It can get to be tricky.

There is now a nice tool to help you edit PC-PATR grammar files. It is called *PcPatrEditor*. Check out the user documentation at https://software.sil.org/downloads/r/paws/pcpatreditor/UserDocumentation.pdf. You can download the latest version at https://software.sil.org/downloads/r/paws/pcpatreditor/PcPatrEditorInstaller.exe.

[5] Some keyboards may not work with this sequence. Therefore we also offer the **Ctrl + L** option.